



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Academic Year: 2023- 2024 (Odd Semester)

Active Learning Activity- Unit-I- Combinational Logic

Degree, Semester & Branch: B.Tech, III Semester & AI&DS -B

Course Code & Title: CS3351 Digital Principles and Computer Organization

Name of the Faculty member: Mrs.V.Krishna Meera

Date: 03.10.2023

Time Allotted for the Activity: 10 minutes

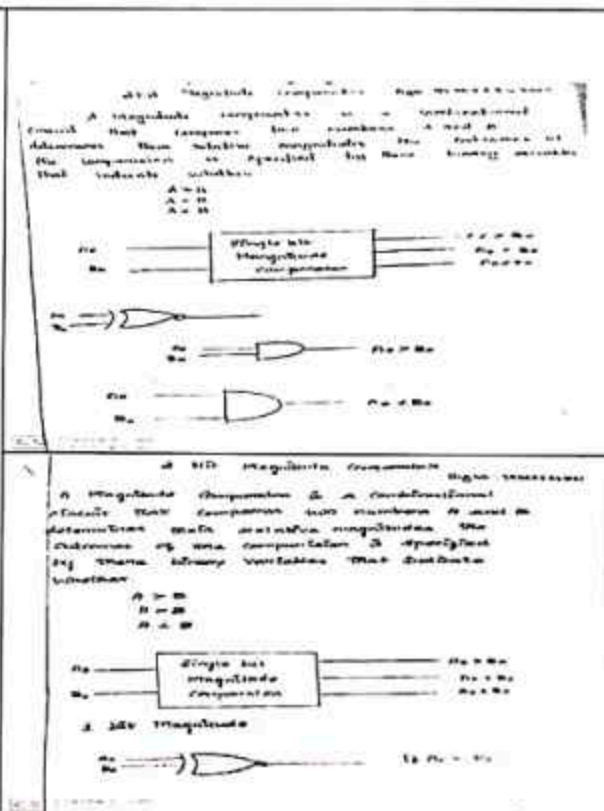
• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1	PSO2
CO1	2	2	2	2	1	1	1	1	1	1

Active Learning Practices Execution

UNIT I - COMBINATIONAL LOGIC

Activity: One Minute Paper
Topic: Magnitude Comparator



V. Krishna Meera
4/10/23
Signature of Faculty Member

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Active Learning Activity- Unit-II- Synchronous Sequential Logic

Degree, Semester & Branch: B.Tech, III Semester & AI&DS -B

Course Code & Title: CS3351 Digital Principles and Computer Organization

Name of the Faculty member: Mrs.V.Krishna Meera

Date: 14.10.202

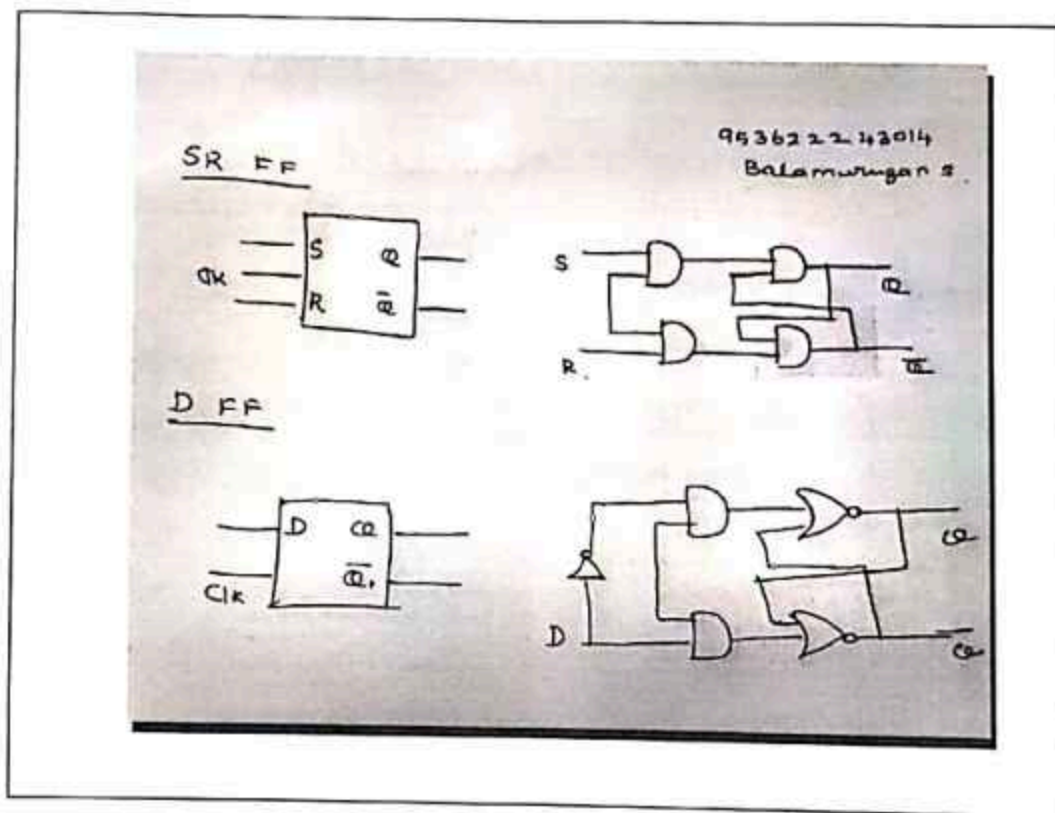
Activity Chosen: Mind Map

Time Allotted for the Activity: 10 minutes

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO12	PSO1	PSO2
CO1	2	2	2	2	1	1	1	1

Flip Flops – SR ,D- Mind map



V. Krishna Meera
15/10/23
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Active Learning Activity- Unit-II- Computer Fundamentals

Degree, Semester & Branch: B.Tech, III Semester & AI&DS -B

Course Code & Title: CS3351 Digital Principles and Computer Organization

Name of the Faculty member: Mrs.V.Krishna Meera

Date: 31.10.2023

Activity Chosen: NPTEL Video

Time Allotted for the Activity: 20 minutes

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	2	2	2	2	1	1	1	1	1	1	1

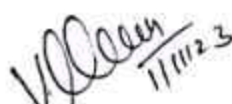
Passive Learning Practices Execution

UNIT III – COMPUTER FUNDAMENTALS

Activity: NPTEL Video

Topic: Fundamental Units Of Digital Computers & Evolution Of Computer




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Department of Electronics and Communication Engineering Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.Tech, III Semester & AI&DS -B

Course Code & Title: CS3351 Digital Principles and Computer Organization

Name of the Faculty member: Mrs.V.Krishna Meera

Date: 31.10.2023

Activity Chosen: Write Pair Share

Time Allotted for the Activity: 40 minutes

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	2	2	2	2	1	1	1	1	1	1	1

Innovative Practice Description

Details of the Implementation:

The objective of the Active Learning Module (ALM) session titled "Write Pair Share" was to enhance students' understanding of Von Neumann Architecture, a fundamental concept in computer science and architecture. The ALM session was conducted on 01/11/2023 in B3L02 and was facilitated by Mrs.V.Krishna Meera. The "Write Pair Share" technique was chosen to promote active engagement and collaborative learning among students.

1. Introduction to Von Neumann Architecture:

- Definition and key components.
- Historical context and significance in computer science.

2. Individual Writing (10 minutes):

- Students were given a set of questions related to Von Neumann Architecture.
- Each student had to independently write their responses, encouraging personal reflection and knowledge recall.

3. Pair Discussion (15 minutes):

- Students were paired up to discuss their written responses.
- Emphasis on comparing and contrasting individual perspectives.
- Encouraged students to articulate their thoughts and question their peers.

4. Group Sharing (20 minutes):

- Each pair shared their key insights with the entire class.
- Facilitator encouraged open discussions, and students were prompted to ask questions or seek clarification.

- Encouraged students to articulate their thoughts and question their peers.
4. Group Sharing (20 minutes):
- Each pair shared their key insights with the entire class.
 - Facilitator encouraged open discussions, and students were prompted to ask questions or seek clarification.
5. Wrap-up and Q&A (5 minutes):
- Summary of the main points covered during the session.
 - Opportunity for students to ask questions and seek clarification on any aspect of Von Neumann Architecture.

Feedback and Observations:

Engagement: The "Write Pair Share" format fostered a high level of engagement among students. The individual writing phase allowed for personal reflection, while the subsequent pair and group discussions promoted collaborative learning.

Inclusivity: The session encouraged the participation of all students, providing a platform for both vocal and non-vocal learners to contribute.

Concept Retention: The majority of students demonstrated a clear understanding of Von Neumann Architecture, as evidenced by their responses during the pair and group sharing phases.

Facilitator's Role: The facilitator played a crucial role in guiding discussions, encouraging critical thinking, and ensuring that all students had the opportunity to express their thoughts.

The ALM "Write Pair Share" session on Von Neumann Architecture was successful in achieving its objective of deepening students' understanding of this foundational concept. The interactive nature of the session promoted active learning and peer-to-peer knowledge exchange. Further, the session provided valuable insights into individual student comprehension and allowed for timely clarification of any misconceptions. Overall, the ALM "Write Pair Share" session was a positive and effective learning experience for students, contributing to a more comprehensive understanding of Von Neumann Architecture.

• Images / Screenshot of the practice:



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11/11/22
Signature of Faculty Member

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HOD



Degree, Semester & Branch: B.Tech, III Semester & AI&DS -B
Course Code & Title: CS3351 Digital Principles and Computer Organization
Name of the Faculty member: Mrs.V.Krishna Meera
Date: 05.12.2023

Activity Chosen: Exit Slips
Time Allotted for the Activity: 20 minutes

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO4	2	2	2	2	2	1	1	1	1	1	1

Questions

Exit Slip

1. Write three things that you learned in today's lesson.
2. Write one question you have about today's lesson.

Images / Screenshot of the practice

- Dissemination
 - Index thousands
 - Dissemination tools covered thousands of documents
 - Indexing
 - From thousands
 - Expanding between the thousands
 - Control thousands
 - Using indexing including dissemination
 - From the thousands
 - Using the thousands by using the thousands
 - From
 - By Technology thousands
 - Thousands
 - Each person concept about the topic

Handed

It is a delay made during execution of instructions

Types of delay

The delay which is caused because of the data dependency control is called

The delay which is caused because the further delay of data

causes

What are the ways to reduce data hazard?

Solutions

1. Data hazard
2. Delay in the dependency of instructions
3. Instruction hazard
4. Delay in the availability of instructions
5. Cache

Temporary storage

A branch penalty can be reduced by branching at the end of the stage

Types of hazards $\rightarrow$ Data, Instruction, Structural

V. J. J. J.
7/1/23

Signature of Faculty Member

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**Department of Electronics and Communication Engineering****Academic Year 2023 – 2024 (Odd Semester)****Degree, Semester & Branch: : III Semester B.Tech.AI&DS-B****Course Code & Title: CS3351 Digital Principles and Computer Organization****Name of the Faculty member (s): Mrs.V.Krishna Meera****Innovative Practice Description**

- **Unit / Topic:** V / Memory and I/O
- **Course Outcome: CO :** The students will be able Identify the characteristics of various memory systems and I/O communication
- **Topic Learning Outcome: TLO :** To explain the different types of memories and its mapping
- **Activity Chosen:** Jigsaw
- **Justification:**
 - The Jigsaw technique is a cooperative learning method that bring about both individual accountability and achievement of the team goals. In this activity students depend on each other to succeed. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal;
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**

The topic is divided into subcategories. Students are divided into groups of five or six students. Then each group would be formed with one student receiving one subcategory of the topic. For this method, each group gets the same set of subcategories. Once individuals have researched and referred their subcategory, they will get together with individuals from the other groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PO12	PSO1
CO5	3	3	2	2	2	2	1

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO8
	3	3	2
Justification for correlation	Basic engineering knowledge is necessary to analyze and characterize the hierarchy of memory. Hence, it is mapped to level 3.	The outcome is mapped to level 3 by identifying the mathematical and engineering knowledge needed	The outcome is mapped to level 2 by demonstrating the types of bus architectures that can be identified by following the



		to solve problems in the memory chip organization.	professional code of ethics while doing activity-based learning.
Innovative practice	PO9	PO10	PSO1
	2	2	1
Justification for correlation	By planning the types of internal communication technologies based on a collaborative activity, Jigsaw helps the students present the results as a team, with smooth integration of contributions from all individual efforts. Hence, the outcome is mapped to level 2.	The topic deals with the design of serial and parallel bus architectures to complement writing and presentations. This task is accomplished through assignments. Hence, it is mapped to level 2.	The students will be able to identify the suitable memory chips for memory allocation. Thus, it is mapped at level 1.

• Images / Screenshot of the practice:



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students gave positive feedback about the activity. They said it was interesting and knowledge improved on the topic. They learn how to communicate with team members and work together.

- ❖ **Benefit of the practice:**

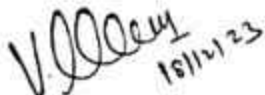
Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 85% of students answered correct

- ❖ **Challenges faced in implementation:**

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ <https://www.readingrockets.org/strategies/jigsaw>
- ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>


Signature of Faculty Member


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